Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “kinetic energy” to explain your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

kinetic energy: energy associated with motion

scalar: a quantity described by size without a vector direction

01 Bring back an earlier idea

An impactor has 3 times the original mass at the same speed and stopping time.
Compare momentum and average stopping-force size.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar.
Neither alone gives a complete damage forecast.

Worked reasoning: A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Two equal-mass carts move at speeds 2 and 4 m/s.
Predict how their momenta and kinetic energies compare before calculating.

Sentence starter: Both … . They differ in … because … .



Same mass; 2 m/s Same mass; 4 m/s

Marzano check - circle: ① ② ③

04 Build the solution

Each cart has mass 2 kg.
Use $K=\frac{1}{2}\times2\times\text{speed}^2$.
Find both kinetic energies.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 24 N net force acts right for 3 s on a 4 kg cart initially at rest.
Find impulse and final velocity.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

For the same mass, speed triples. By what factor does kinetic energy change?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Find two carts with equal mass but speeds in a 1: 2 ratio.
Compare momentum and energy multipliers.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3